机器学习第八次作业

Problem 1: Using the axioms of probability, prove

$$p(P \mid P \wedge Q) = 1$$

$p(P|P \wedge Q) = \frac{p(P \wedge (P \wedge Q))}{p(P \wedge Q)}$。
根据逻辑幂等律和结合律，$P \wedge (P \wedge Q)$等价于$P \wedge Q$。
因此：$p(P|P \wedge Q) = \frac{p(P \wedge Q)}{p(P \wedge Q)} = 1$。

Problem 2: Given the Bayesian Network

- Calculate $p(P \Rightarrow Q)$

$= p(\neg P \vee Q) = p(\neg P) + p(Q) - p(\neg P \wedge Q) = p(\neg P) + p(P \wedge Q)$

- In what condition, the following relation is held

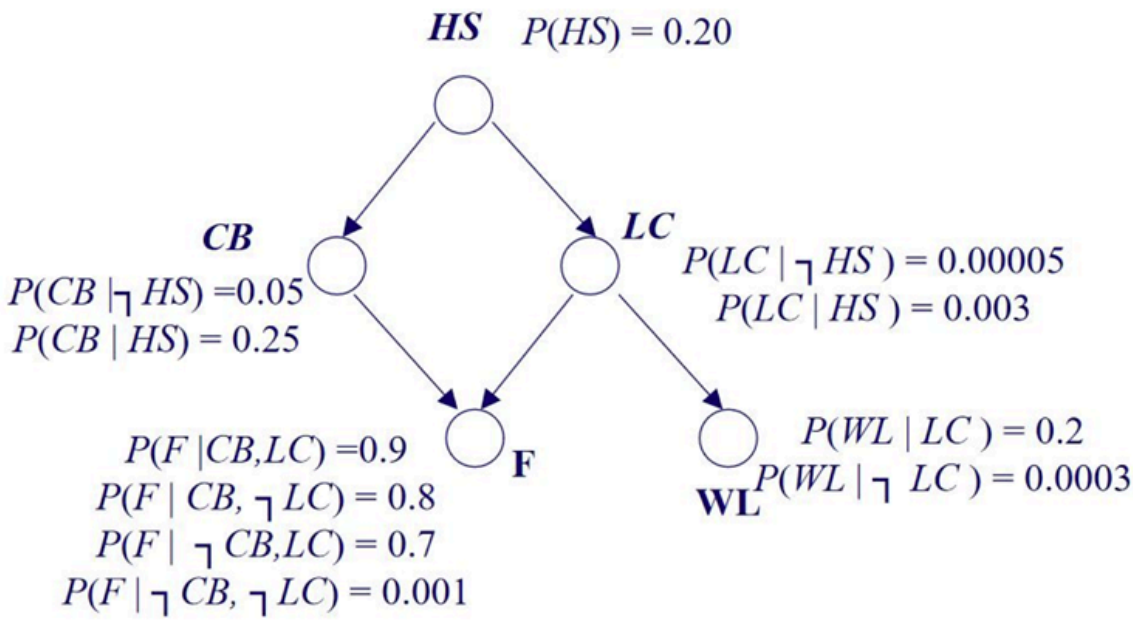
$$p(P \Rightarrow Q) = p(Q|P)$$

$p(\neg P) + p(P \wedge Q) = \frac{p(P \wedge Q)}{p(P)}$
求解得到：
$(p(\neg P))(p(P) - p(P \wedge Q)) = 0$

则需要满足下面两个条件其中之一：

- $p(P) = 1$: 此时$p(Q|P) = p(Q)$, 且$p(P \Rightarrow Q) = p(Q)$。
- $p(P \Rightarrow Q) = 1$

Problem 3： Given the following Bayesian Network, calculate $p(LC \mid F, WL)$



$$p(LC|F,WL) = \frac{p(LC,F,WL)}{p(F,WL)} = \frac{p(LC,F,WL)}{p(LC,F,WL) + p(\neg LC,F,WL)}$$

计算$p(LC)$：

$$\begin{aligned} p(LC) &= p(LC|HS)p(HS) + p(LC|\neg HS)p(\neg HS) \\ &= (0.003 \times 0.20) + (0.00005 \times 0.80) \\ &= 0.0006 + 0.00004 = 0.00064 \\ p(\neg LC) &= 1 - 0.00064 = 0.99936 \end{aligned}$$

计算$p(CB)$：

$$\begin{aligned} p(CB) &= p(CB|HS)p(HS) + p(CB|\neg HS)p(\neg HS) \\ &= (0.25 \times 0.2) + (0.05 \times 0.8) = 0.09 \\ p(\neg CB) &= 0.91 \end{aligned}$$

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\begin{aligned}
p(F|LC) &= p(F|CB, LC)p(CB) + p(F|\neg CB, LC)p(\neg CB) \\
&= (0.9 \times 0.09) + (0.7 \times 0.91) \\
&= 0.081 + 0.637 = 0.718 \\
p(F|\neg LC) &= p(F|CB, \neg LC)p(CB) + p(F|\neg CB, \neg LC)p(\neg CB) \\
&= (0.8 \times 0.09) + (0.001 \times 0.91) \\
&= 0.072 + 0.00091 = 0.07291
\end{aligned}
$$

代入最后结果：

$p(LC, F, WL) = p(F|LC) \times p(WL|LC) \times p(LC) = 0.718 \times 0.2 \times 0.00064 \approx 0.0000919$

$p(\neg LC, F, WL) = p(F|\neg LC) \times p(WL|\neg LC) \times p(\neg LC) = 0.07291 \times 0.0003 \times 0.99936 \approx 0.0000219$

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p(LC|F, WL) = \frac{0.0000919}{0.0000919 + 0.0000219} \approx 0.807
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